

# COMP3162-LAB02-620043505

2025-02-03

## Lab 2

### Question 1

Mean, mode and median values for:

a) Sepal.Length

```
mean(iris$Sepal.Length)
```

```
## [1] 5.843333
```

b) Sepal.Width

```
mean(iris$Sepal.Width)
```

```
## [1] 3.057333
```

c) Petal.Length

```
mean(iris$Petal.Length)
```

```
## [1] 3.758
```

d) Petal.Width

```
mean(iris$Petal.Width)
```

```
## [1] 1.199333
```

### Question 2

What is the minimum.

a) Sepal.Length

```
min(iris$Sepal.Length)
```

```
## [1] 4.3
```

b) Sepal.Width

```
min(iris$Sepal.Width)
```

```
## [1] 2
```

### Question 3

What is the largest:

a) Petal.Length

```
max(iris$Petal.Length)
```

```
## [1] 6.9
```

a) Petal.Width

```
max(iris$Petal.Width)
```

```
## [1] 2.5
```

### Question 4

Using the data in the column named Species, determine:

a) How many species exist in the dataset? (hint: `summary()` and `length()` functions can be used)

Species Existing:

```
summary(iris$Species)
```

```
##      setosa versicolor  virginica  
##       50         50         50
```

Total Count:

```
length(iris$Species)
```

```
## [1] 150
```

## Question 5

Analysis of the Versicolor Species:

- a) Extract all data for the ‘versicolor’ species and place it in a new object called v.species. (*Hint: use a conditional statement between the square brackets to select the rows that have the matching Species value. Leave the column value empty. An example is in the lecture notes*)

```
v.species <- iris[iris$Species == "versicolor",]  
head(v.species)
```

```
##      Sepal.Length Sepal.Width Petal.Length Petal.Width   Species  
## 51          7.0         3.2         4.7         1.4 versicolor  
## 52          6.4         3.2         4.5         1.5 versicolor  
## 53          6.9         3.1         4.9         1.5 versicolor  
## 54          5.5         2.3         4.0         1.3 versicolor  
## 55          6.5         2.8         4.6         1.5 versicolor  
## 56          5.7         2.8         4.5         1.3 versicolor
```

- b) How many of the v.species have both a Sepal.Length and Petal.Length greater than 4? (*you will need to use the hint from part 5a to help you here in addition, consider the nrow() and ncol() functions*).

```
nrow(v.species[v.species$Sepal.Length > 4 & v.species$Petal.Length > 4,])
```

```
## [1] 34
```

## Question 6

Several variations exist between sepals and petals:

- a) Add a new column called avg\_dist to the iris object which has the *average distance* for each row as calculated (*Ensure you show the result in your rmarkdown file*). The *average distance* for a row is calculated as:

$$((\text{Sepal.Width} - \text{Petal.Width}) + (\text{Sepal.Length} - \text{Petal.Length}))/2$$

```
iris$avg_dist <- ((iris$Sepal.Width - iris$Petal.Width) + (iris$Sepal.Length - iris$Petal.Length))/2
```

- b) What is the average distance between the sepal and petals in this species (this is the average of all rows).

```
mean(iris$avg_dist)
```

```
## [1] 1.971667
```

## Question 7

For each species, what is the mean *average distance*? (*requires calculating for each species separately*)

```
aggregate(iris$avg_dist, by = list(iris$Species), FUN = mean)
```

```
##      Group.1      x
## 1      setosa 3.363
## 2 versicolor 1.560
## 3  virginica 0.992
```

## Question 8

Add another column to the dataset called `s.group` which sets each value to either: 0 or 1 based on whether the value in the `avg_dist` column. If the value of `avg_dist` is less than or equal to the *mean average distance*, then the current row in `s.group` should be set it to 0, otherwise set it to 1. (*hint: you can use the `ifelse` function*)

```
iris$s.group <- ifelse(iris$avg_dist <= mean(iris$avg_dist), 0, 1)
head(iris)
```

| ##   | Sepal.Length | Sepal.Width | Petal.Length | Petal.Width | Species | avg_dist | s.group |
|------|--------------|-------------|--------------|-------------|---------|----------|---------|
| ## 1 | 5.1          | 3.5         | 1.4          | 0.2         | setosa  | 3.50     | 1       |
| ## 2 | 4.9          | 3.0         | 1.4          | 0.2         | setosa  | 3.15     | 1       |
| ## 3 | 4.7          | 3.2         | 1.3          | 0.2         | setosa  | 3.20     | 1       |
| ## 4 | 4.6          | 3.1         | 1.5          | 0.2         | setosa  | 3.00     | 1       |
| ## 5 | 5.0          | 3.6         | 1.4          | 0.2         | setosa  | 3.50     | 1       |
| ## 6 | 5.4          | 3.9         | 1.7          | 0.4         | setosa  | 3.60     | 1       |